

Finite Markov Networks: Matrix Trees, Entropy Production, and Exact Fluctuation Symmetry

HCS-C361 theorem package

4 September 2026

Abstract

For every finite irreducible continuous-time Markov chain with bidirected support, we give a convention-complete proof joining the matrix-tree stationary law, positivity and exact equality criteria for entropy production, a finite-time stationary path-reversal theorem, and transpose symmetry of the entire tilted characteristic polynomial. Total and medium entropy are kept distinct. Rate-function symmetry is stated only under explicit large-deviation and Legendre hypotheses. Exact rational ledgers guard all orientation and sign conventions; they do not replace the analytic proof.

1 Frozen chain and theorem

Let $S = \{0, \dots, d-1\}$ be finite. For $i \neq j$, assume $q_{ij} > 0$ iff $q_{ji} > 0$, and that the undirected support is connected. There are no recorded self-jumps. On column functions,

$$(Lf)(i) = \sum_{j \neq i} q_{ij}(f(j) - f(i)), \quad L_{ij} = q_{ij}, \quad L_{ii} = -r_i, \quad r_i = \sum_{j \neq i} q_{ij}.$$

Let $T \rightarrow i$ denote a directed spanning tree whose edges point toward root i , and set $\tau_i = \sum_{T \rightarrow i} \prod_{(u \rightarrow v) \in T} q_{uv}$.

Theorem 1 *The unique stationary row law is $\pi_i = \tau_i / \sum_k \tau_k$. With*

$$a_{ij} = \pi_i q_{ij}, \quad J_{ij} = a_{ij} - a_{ji}, \quad F_{ij} = \log(a_{ij}/a_{ji}),$$

the entropy-production rate satisfies

$$\sigma = \sum_{i < j} (a_{ij} - a_{ji}) \log \frac{a_{ij}}{a_{ji}} \geq 0.$$

Moreover, $\sigma = 0$, detailed balance, and zero affinity $\sum_{(i,j) \in C} \log(q_{ij}/q_{ji}) = 0$ on every oriented cycle C are equivalent. Start in π . For a path with state skeleton $i_0, \dots, i_m$, define

$$\Sigma_T = \log \frac{\pi_{i_0} \prod_{\ell=1}^m q_{i_{\ell-1} i_\ell}}{\pi_{i_m} \prod_{\ell=1}^m q_{i_\ell i_{\ell-1}}}.$$

If Θ reverses time, define the reversed pushforward law $\mathbb{P}_\pi^R = \mathbb{P}_\pi \circ \Theta^{-1} = \mathbb{P}_\pi \circ \Theta$. Then

$$\frac{d\mathbb{P}_\pi}{d\mathbb{P}_\pi^R} = e^{\Sigma_T}, \quad \mathbb{E} e^{-\Sigma_T} = 1, \quad \mathbb{E} e^{-\lambda \Sigma_T} = \mathbb{E} e^{-(1-\lambda)\Sigma_T}.$$

More precisely, its law μ_T obeys $\mu_T(B) = \int_{-B} e^{-s} \mu_T(ds)$, where $-B = \{-s : s \in B\}$; a point-probability formula is intended only at atoms. The exponential identity holds for every real λ .

For medium entropy $W_T = \sum \log(q_{ij}/q_{ji})$, the tilted matrix is

$$(L_\lambda)_{ij} = q_{ij}^{1-\lambda} q_{ji}^\lambda \quad (i \neq j), \quad (L_\lambda)_{ii} = -r_i.$$

It satisfies $L_\lambda^\top = L_{1-\lambda}$ and hence, for all $z \in \mathbb{C}$,

$$\det(zI - L_\lambda) = \det(zI - L_{1-\lambda}).$$

For every real λ , the SCGF $\psi(\lambda) = \lim_{T \rightarrow \infty} T^{-1} \log \mathbb{E}_\pi e^{-\lambda W_T}$ exists, equals the Perron eigenvalue of L_λ , is finite and real analytic, and has the same symmetry. If, additionally, W_T/T obeys a full LDP whose rate equals the Legendre–Fenchel transform $I(a) = \sup_\lambda \{-\lambda a - \psi(\lambda)\}$, then $I(a) - I(-a) = -a$.

2 Proof

For $M = -L^\top$, principal-cofactor expansion cancels selections containing directed cycles and leaves precisely the in-arborescence products. Thus $L^\top \tau = 0$. Connectivity gives $\tau_i > 0$ for every root; irreducibility makes the nullspace one-dimensional, proving the stationary formula.

The inequality $(x - y) \log(x/y) \geq 0$ for $x, y > 0$ proves nonnegativity edge by edge, with equality exactly when $a_{ij} = a_{ji}$. Detailed balance telescopes around every cycle. Conversely, assume all cycle products are one. Fix o and define along any support path

$$h_i = \prod_{o=i_0 \rightarrow \dots \rightarrow i_m=i} \frac{q_{i_{\ell-1}i_\ell}}{q_{i_\ell i_{\ell-1}}}.$$

The value is path-independent: concatenate two paths, erase backtracks, and split at repeated vertices into simple cycles. Therefore $h_j/h_i = q_{ij}/q_{ji}$ on every edge. Normalized h satisfies detailed balance and stationarity, so it equals π .

For fixed jump skeleton and holding intervals $s_0, \dots, s_m$, the path density equals

$$\pi_{i_0} \exp\left(-\sum_{\ell=0}^m r_{i_\ell} s_\ell\right) \prod_{\ell=1}^m q_{i_{\ell-1}i_\ell}.$$

Reversal only reorders the holding exponential and reverses each jump. Bidirectionality therefore gives the stated Radon–Nikodym derivative against $\mathbb{P}_\pi^R$, including zero-jump paths, and $\Sigma_T \circ \Theta = -\Sigma_T$. Change of variables yields $\mathbb{E}[g(\Sigma_T)] = \mathbb{E}[e^{-\Sigma_T} g(-\Sigma_T)]$; indicators and exponentials give every finite-time identity.

Feynman–Kac multiplication of a jump $i \rightarrow j$ by $e^{-\lambda \log(q_{ij}/q_{ji})}$ gives the displayed tilt, whose transpose identity is entrywise. Transposition preserves the full characteristic polynomial. After a scalar shift, Perron–Frobenius supplies a simple real dominant eigenvalue and finite-dimensional Feynman–Kac identifies it with ψ . Since the tilted entries depend analytically on real λ and this eigenvalue is simple, the local branches join to make ψ real analytic on $\mathbb{R}$. Finally, $|\Sigma_T - W_T| \leq 2 \max_i |\log \pi_i|$, so total and medium entropy have the same SCGF. Substitution $\mu = 1 - \lambda$ in the explicitly assumed Legendre formula proves the rate symmetry; no rate-function conclusion is drawn without that full-LDP identification.

3 Sharp boundaries and a minimal nonequilibrium cycle

For one state, the empty tree has weight one and all entropy is zero. Every irreducible two-state chain is reversible because stationarity forces $\pi_0 q_{01} = \pi_1 q_{10}$; indeed $\Sigma_T = 0$ pathwise. A nonzero affinity first occurs with three states. Give every clockwise edge rate 2 and every reverse edge rate 1. Then $\pi_i = 1/3$, the clockwise affinity is $3 \log 2$, each oriented stationary current is $1/3$, and $\sigma = \log 2$.

Reducibility destroys the unique positive global law and positive rooted-tree weights for every root; the theorem applies after selecting a closed irreducible class. A one-way edge can make the reversed path measure singular and the affinity infinite, so it is not a zero-rate substitution into this theorem. Phantom self-transitions are excluded; diagonals encode holding.

4 Exact receipts, sources, and scope firewall

Four frozen rational panels have sizes two through five. The canonical artifact enumerates 700 directed trees, all canonical simple cycles and unordered edges, 24 tilted characteristic polynomials, 2,244 path skeletons, and five boundary rows. The producer uses tree enumeration and permutation expansion; the independent checker uses Laplacian cofactors and Faddeev–LeVerrier. SymPy is a third lane. Repaired-hash attacks target schemas, coordinates, rows, evaluation semantics, and strict parsers. These are regression receipts, not a finite proof of Theorem 1.

Schnakenberg’s network theory (Rev. Mod. Phys. 48 (1976), DOI 10.1103/RevModPhys.48.571) supplies the graph-thermodynamic lineage. Lebowitz and Spohn (J. Stat. Phys. 95 (1999), DOI 10.1023/A:1004589714161) supply the stochastic-action and fluctuation-theorem lineage. We claim no priority and rederive every statement under the frozen convention.

The tilted determinant is only a finite Markov-network characteristic polynomial. It is not a target Euler factor, target functional equation, arithmetic primitive-orbit zeta, root number, divisor, or Hilbert–Pólya operator. All Route-A gates fail; Route B is not invoked. Scope is NO_BAD_EULER_OR_ROOT_NUMBER.